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GitOps on OpenShift

Declarative Operations from Git

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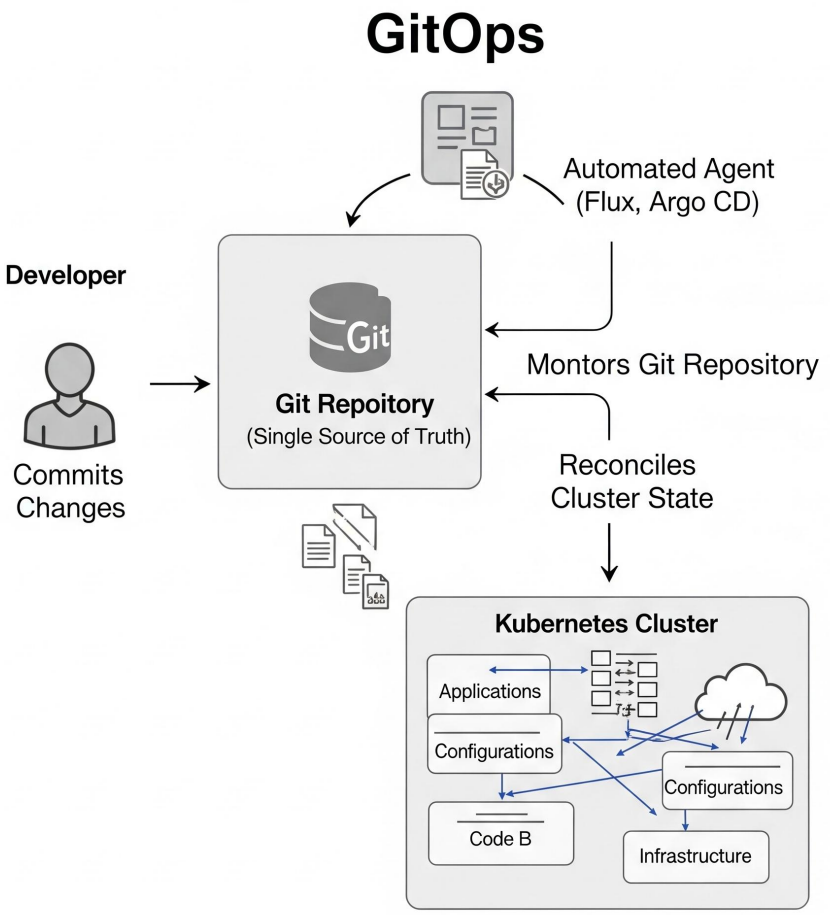
Agenda

- What is GitOps and why it matters
- GitOps on OpenShift – options
- OpenShift GitOps (Argo CD)
- Key concepts: Applications, AppProject, ApplicationSet
- GitOps vs CI/CD pipelines
- Security and multi-tenancy
- Integration with ACM and policies
- Best practices and summary



GitOps Fundamentals

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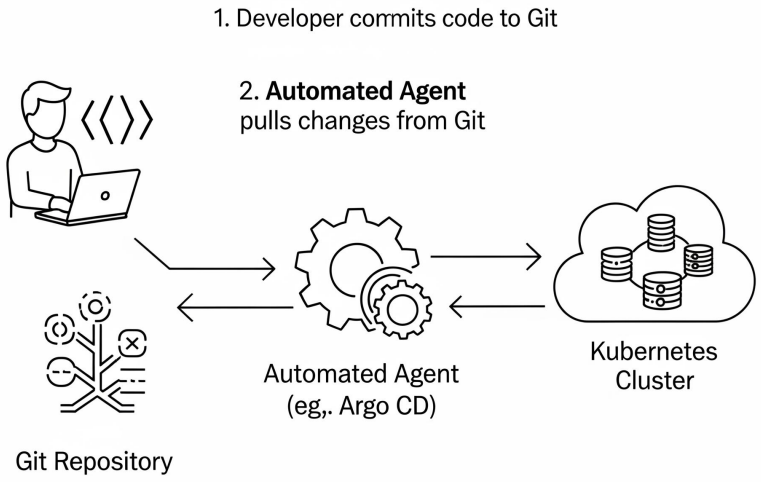
What Is GitOps

Operations as code

- Desired state lives in Git – single source of truth
- Automated agents reconcile cluster to match Git
- Pull-based: cluster pulls changes; no direct kubectl in prod
- Auditable history, rollbacks via revert commits
- Aligns with Kubernetes declarative APIs

GitOps Flow



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Why GitOps on OpenShift

Enterprise Kubernetes

- Many teams and namespaces – Git enforces consistency
- Supports compliance: who merged what, when
- Works with OpenShift resources (Routes, BuildConfigs optional)
- Red Hat supports OpenShift GitOps operator (Argo CD)
- Scales to multi-cluster with Advanced Cluster Management

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Options on OpenShift

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GitOps Tooling Landscape

OpenShift ecosystem

- OpenShift GitOps:
- Argo CD–based operator
- Red Hat supported
- UI in console integration
- Default choice for many customers

- Alternatives:
- Flux (GitOps Toolkit)
- Pipelines + GitOps hybrid
- ACM GitOps for multi-cluster
- Pick one pattern per org standard

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Standardize on one GitOps controller per cluster or fleet.

OpenShift GitOps (Argo CD)

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OpenShift GitOps Operator

Installation and lifecycle

- Install from OperatorHub – OpenShift GitOps operator
- Deploys Argo CD in a namespace (often openshift-gitops)
- Managed updates aligned with OpenShift releases
- Integrates with OpenShift OAuth for console SSO
- Use for application manifests and optional platform config

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Core Argo CD Concepts

What you configure

- Application: links a Git repo path to a destination cluster/namespace
- AppProject: boundaries, allowed repos, destinations, roles
- ApplicationSet: generate many Applications from generators
- Sync policy: manual vs automated, prune, self-heal
- Health and sync status drive UI and alerts

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Git Repository Layout

Mono-repo vs multi-repo

- App repo: manifests per microservice or per environment
- Kustomize or Helm for overlays (dev/stage/prod)
- Platform repo: cluster config, operators, policies (optional GitOps)
- Separate secrets: Sealed Secrets, External Secrets, or Vault
- Branch or directory per environment – team convention

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GitOps and CI/CD

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GitOps vs CI/CD

Build vs deploy

- CI: build, test, scan, push image to registry
- CD (GitOps): update manifests or image tags in Git; Argo syncs
- Tekton / Jenkins for CI; Argo CD for continuous deployment
- Webhook or pipeline step commits manifest changes
- GitOps does not build images – it deploys what Git describes

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Security and Governance

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Multi-Tenancy and RBAC

Shared Argo CD

- Use AppProjects to isolate teams and allowed namespaces
- Map OpenShift groups to Argo CD RBAC via SSO
- Limit which Git repos each project can use
- Separate Argo instances for platform vs apps if needed
- Audit sync events and failed health checks

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Policies and ACM

Beyond a single cluster

- Advanced Cluster Management can deploy policies and GitOps subscriptions
- Align cluster configuration with organizational standards
- Use Policy-as-code to enforce labels, operators, or security
- GitOps ApplicationSet can target multiple clusters
- Single Git workflow for many OpenShift clusters

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Best Practices

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GitOps Best Practices

Production readiness

- Start with non-prod; prove sync and rollback
- Enable self-heal cautiously; understand drift vs hotfixes
- Use progressive delivery (Argo Rollouts) when needed
- Monitor Argo CD as a critical control plane
- Document promotion: dev → stage → prod via Git branches or paths

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Summary

GitOps on OpenShift

- GitOps: declarative desired state from Git, continuous reconciliation
- OpenShift GitOps delivers supported Argo CD on the platform
- Application, AppProject, ApplicationSet structure your rollout
- Pair with CI for images; use secrets operators for sensitive data
- Scale with ACM and policies for multi-cluster GitOps

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Thank You

Questions? See docs.redhat.com – OpenShift GitOps.

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